



Research Paper

Corruption culture of multinationals: Evidence from China

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ABSTRACT

Using administrative data and tax returns from multinational corporations, we investigate the impact of foreign cultural norms on corrupt practices within China. By leveraging time-varying ownership details of foreign shareholders, our research reveals that multinationals originating from countries with a prevalent corruption culture tend to exhibit greater corruption in China, as evidenced by increased entertainment costs, with a pronounced effect in industries with more rent-seeking activities. Corrupt foreign firms are more likely to secure subsidies and tax refunds from local governments and realize higher profitability. However, these practices lead to an insignificant change in their innovation outcomes. China's anti-corruption campaign has significantly reduced the influence of foreign corruption culture, highlighting the importance of robust regulatory enforcement in shaping the behavior of foreign firms operating in emerging economies. Additionally, the influence of foreign culture extends to domestic firms, demonstrating a cultural spillover effect.

1. Introduction

Do multinational corporations bribe in emerging markets and what are the implications? Corruption has been identified as “public enemy number one”, a sentiment expressed by World Bank Group President Jim Yong Kim in 2013.¹ In the context of an increasingly globalized economy, corruption has transcended local boundaries and has emerged as a global and complex problem. Multinational corporations (MNCs) are central to this phenomenon, as they often operate across multiple legal and cultural environments, creating opportunities for corrupt practices. High-profile cases involving MNCs, such as Walmart's \$24 million bribery of Mexican officials in the early 2000s, Siemens AG and Pfizer's bribery in China, and Goldman Sachs Group's recent over \$1 billion bribes in Malaysia, underscore the significant role these entities play in international corruption (Barstow, 2012; Rose-Ackerman and Tan, 2014; U.S. Department of Justice, 2020). The magnitude of this problem is alarming, with estimates suggesting that bribes paid by western MNCs have already reached \$80 billion per year around 2000, a sum comparable to the United Nations' estimated resources required to eradicate global poverty (The Corner House, 2000).

The widespread engagement of MNCs in corrupt activities, particularly in emerging markets, has drawn substantial academic interest in understanding the underlying mechanisms driving these behaviors. While existing literature has examined factors such as competition, institutional frameworks, and agency problems in explaining MNCs' corruption activities in host countries (see recent reviews by Cuervo-Cazurra, 2016; Bahoo et al., 2020), the influence of cultural norms from MNCs' home countries has been largely overlooked. MNCs, influenced by the cultural norms of their home countries, encounter diverse ethical standards and cultural environments when entering new markets. These foreign investors could transfer their cultural imprints to the host country, influencing their behaviors in ways that reflect the norms of their origin. Although recent studies have highlighted the importance

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¹ Details [here](#).

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0167-2681/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

of cultural norms from the home country in shaping MNC behavior in areas such as gender equality, firm transparency, and tax evasion (Braguinsky and Mityakov, 2015; DeBacker et al., 2015; Tang and Zhang, 2021), the specific impact of these norms on corrupt practices has not been thoroughly investigated.

This study seeks to fill this gap by investigating the influence of the home country's corruption culture on the corruption behaviors of MNCs operating in China. In particular, we utilized administrative data on the ownership of foreign shareholders to examine the relationship between foreign firms' corruption behaviors in China and the corruption culture of their home countries. This setting levels foreign firms with varying cultural norms from home country within the same regulatory and cultural environment in China, and allows us to isolate the effect of cultural norms on corruption behavior.

Measuring corruption poses significant challenges due to its covert and illegal nature. Prior research has utilized observable indicators such as entertainment and travel expenses, to infer corrupt activities (Cai et al., 2011; Huang et al., 2017; Fang et al., 2023). Building on this methodology, we refine this approach using detailed firm-level data on entertainment costs (EC) to measure corruption behaviors. We validate this measure by demonstrating its significant positive correlation with other survey-based and aggregate measures of corruption. Additionally, we quantify firm-level exposure to corruption culture by calculating a weighted average of the Corruption Perceptions Index (CPI) of the investors' home countries, with weights based on the ownership share of each investor. The CPI, developed by Transparency International, is widely recognized as a reliable measure of corruption culture (DeBacker et al., 2015; Asteriou et al., 2021).

Our baseline findings reveal that multinationals with a prevalent corruption culture in their home countries tend to exhibit greater corruption behaviors in China. This relationship persists across various model specifications, variable measurements, and sampling methods. When firm- and year-fixed effects are included, one standard deviation increase in 2.376% increase relative to the mean of the entertainment cost, suggesting an economically sizable impact. The effect is also more pronounced in high-rent-seeking industries, where monopoly rents are high and government influence is substantial.

Corrupt activities have tangible economic consequences. Our findings indicate that MNCs with larger ECs are more likely to receive higher subsidies and tax refunds from Chinese governments, and achieve greater profitability. However, there is no significant correlation between increased ECs and MNCs' innovation activities, suggesting that the enhanced government benefits and profitability have not translated into fostering innovation, an essential driver of long-term economic growth. Furthermore, considering that MNCs exhibit elevated levels of corrupt practices, particularly in high rent-seeking industries, these results raise concerns about the economic costs of corruption.

Considering the above economic consequences, how can the corrupt culture transmission and corrupt activities of MNCs in the host country be curtailed? It has long been recognized that cultural norms and legal enforcement are both important determinants of corruption (Fisman and Miguel, 2007). The anti-corruption program is an important way of enforcement to fight corruption (Colonnelli and Prem, 2022). China sets out the anti-corruption program after the 18th National Congress of the Communist Party in 2012, and the coverage and impact of this anti-corruption campaign are broader and more lasting than before. Our findings indicate that China's anti-corruption program has successfully suppressed the influence of foreign corruption culture on MNCs' behavior. Although the cultural effect on corruption was significantly positive before the program's implementation, it was quickly curtailed and remains suppressed even four years after the program's launch. We also find that a better local legal environment can limit the corruption culture's effect of MNCs on their local corrupt activities.

More surprisingly, we observe that MNCs also influence the corruption behaviors of domestic firms, with domestic firms' corruption levels reducing in the presence of foreign affiliates from less corrupt cultures in the same prefecture. This finding demonstrates a broader cultural spillover effect that extends beyond the MNCs themselves.

Our study contributes to extensive research on culture and firms, most of which focuses on the effect of culture on local firms' performances (e.g. Guiso et al., 2004, 2006; Ahern et al., 2015; García-Vega and Huergo, 2017; Kong et al., 2021; Luo et al., 2024). An emerging strand of studies in this research begins to investigate how cultural norms impact financial outcomes across countries, such as cross-border investments (Guiso et al., 2009; Hwang, 2011; Ahern et al., 2015; Bottazzi et al., 2016), interest rates (Giannetti and Yafeh, 2012), stock market participation, and stock price momentum (Chui et al., 2010). We are more aligned with this literature in investigating the cultural norms impact on MNCs' cross-country behaviors. Yet, we contribute to this literature by studying the specific cultural norms of corruption and the behaviors of corruption in the host economy. Our study is particularly related to DeBacker et al. (2015) in this literature, which examines how corruption norms in foreign countries affect tax evasion by MNCs in the United States. While both studies focus on the impact of foreign culture on MNCs' behaviors in host countries, our research diverges in three key ways: (a) We focus on China, the largest emerging economy and one of the world's biggest recipients of MNC investments, where corruption issues are especially pertinent. In contrast, DeBacker et al. (2015) study firm behaviors in the United States, the most developed economy. (b) Our study directly addresses the corruption activities of MNCs in China, whereas DeBacker et al. (2015) focus on tax evasion. (c) We leverage detailed, time-varying ownership data of foreign shareholders in China to construct a firm-level measure of exposure to foreign cultural norms, while they use a single foreign country's culture as the basis for their analysis.

Our study also contributes to the vast literature on corporate corruption behavior (e.g. Shleifer and Vishny, 1993; Svensson, 2003, 2005; Cheung et al., 2006, 2021; Fisman and Svensson, 2007; Olken, 2007; Olken and Pande, 2012; Alm et al., 2016; Litina and Palivos, 2016; Liu, 2016; Boly and Gillanders, 2018; Bai et al., 2019; Le et al., 2020). While our results regarding the benefits

of corruption echo this literature, we contribute by documenting how home-country corruption cultural values influence MNCs' corrupt behaviors.²

Moreover, our findings on the spillover effects of MNCs' corruption behaviors resonate with the broader literature on MNCs' spillover effects on local firms. While the classic view emphasizes the spillover of technology from MNCs to local firms (e.g. Aitken and Harrison, 1999; Javorcik, 2004; Liu et al., 2024), an emerging strand of research explores the cultural spillover effects of MNCs on local firms' behaviors, such as MNCs' gender equality culture on local firms' employment of female managers (Tang and Zhang, 2021) and female education (Guo et al., 2023); the MNC transparency culture on the transparency of local firms' operations (Braguinsky and Mityakov, 2015), and the impact of MNCs' corruption culture on local firms tax avoidance behaviors (DeBacker et al., 2015). In addition, our analysis of the anti-corruption crackdown in China also complements the literature on the legal enforcement of corruption behavior, such as Fisman and Miguel (2007) and Colonnelli and Prem (2022).

2. Hypothesis development

Cultural norms are profound imprints that shape the behavior of both individuals and organizations. These unspoken rules and expectations for acceptable conduct are learned through socialization and become internalized as personal and organizational values that guide actions and decisions (Hofstede, 2001; Guiso et al., 2004, 2006; Fisman and Miguel, 2007; Hofstede et al., 2010). Multinational corporations (MNCs) are influenced by the cultural norms of their home countries and encounter a range of ethical standards and cultural environments when entering new markets. As foreign investors, they often bring their own cultural practices with them, which can shape their behaviors in ways that reflect their origins. However, they can also be affected by the local cultural norms of the host country, which may sometimes conflict with the values and practices established in their home countries.

Corruption culture plays an important role in shaping the behaviors of individuals and firms, particularly when it comes to illegal activities (e.g. Fisman and Miguel, 2007; DeBacker et al., 2015). Will the corruption culture of MNCs' home country influence their corruption behaviors in the host economy? Some argue that foreign investment is positively associated with corruption, especially in emerging countries. A subsidiary of an MNC might seek external legitimacy by engaging in common corruption practices within the host economy or resorting to bribery to address the competitive disadvantages experienced by foreign companies in those countries (Hellman et al., 2000, 2002; Kwok and Tadesse, 2006; Zhu, 2017; Yi et al., 2018). However, some scholars argue that foreign investment can reduce corruption. In addition to the external pressures from the host economy, subsidiaries of MNCs also face significant influences from their headquarters in the home-country environment, which may have norms and practices to restrain corruption behaviors (Kwok and Tadesse, 2006). MNCs may also increase competition and diffuse best practices in corporate governance to reduce corruption (Kwok and Tadesse, 2006; Jiang et al., 2024). Therefore, we have our following hypothesis:

H1: Foreign corruption culture is positively correlated with the corruption behavior of multinationals in China.

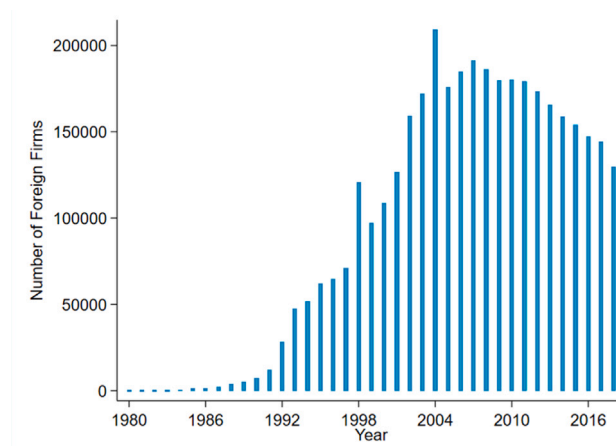
While MNCs can bring capital, technology, and other benefits to the host economy, the transmission of a corruption culture from parent countries to their affiliates may undermine some of these advantages. To address this challenge, it is crucial to consider effective strategies that host countries can implement to restrain the corruption culture emanating from parent countries. The World Bank highlights the effectiveness of legal enforcement in combating corruption. Effective legal frameworks and enforcement mechanisms are essential in deterring corrupt practices. Strengthening anti-corruption laws and regulations, ensuring their consistent application, and prosecuting offenders at all levels can send a strong message and curb corruption (Fisman and Miguel, 2007; Chen and Kung, 2019; Colonnelli and Prem, 2022). Therefore, the rigorous enforcement of anti-corruption laws and programs in the host country can help mitigate the positive correlation between the home culture and the corrupt practices of MNCs operating in the host country. Consequently, we propose our second hypothesis.

H2: Enforcement of anti-corruption laws and programs can curb the transmission of corruption culture from home countries to affiliates in China.

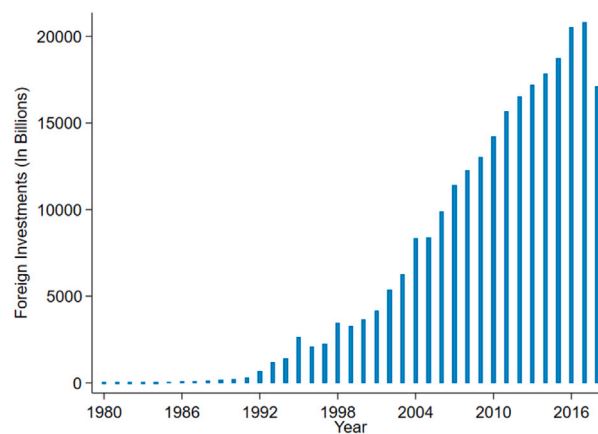
MNCs can significantly influence the establishment of organizational frameworks within host countries (Westney, 1993). They introduce new business practices that could challenge existing norms and stimulate discussions on improved business conduct (Kwok and Tadesse, 2006). This "demonstration effect" can lead to a reduction in corruption among local firms, particularly when MNCs operate with lower levels of corruption themselves. In addition, MNC employees play a crucial role in transferring corporate governance and cultural practices to local firms (Zhang et al., 2018). By cultivating a skilled labor pool in the host economy, these employees can bring valuable insights and practices to local companies when they transition to those firms, thereby enhancing corporate governance and ethical standards in the local business environment. We develop our third hypothesis below.

H3: MNCs from less corrupt countries will transfer culture to local firms and reduce local firms' corruption behaviors.

² There is also one literature examining the influence of foreign ownership on corruption. This body of research generally addresses foreign ownership broadly, without delving into the specific cultural norms of MNCs. Some scholars argue that foreign investment can reduce corruption by increasing competition and diffusing best practices in corporate governance (Kwok and Tadesse, 2006; Jiang et al., 2024). These findings align with our spillover results when foreign capital originates from less corrupt cultures. Conversely, other studies have found a positive relationship between foreign investment and corruption, suggesting that bribery may be used to overcome competitive disadvantages faced by foreign firms in host countries (Hellman et al., 2000, 2002; Malesky et al., 2015; Zhu, 2017; Yi et al., 2018). The lack of consensus in this literature highlights the need to uncover the specific mechanisms at play within foreign ownership, which we address by focusing on the cultural norms dimension.



(a) Number of Foreign Firms in China



(b) Total Equity Investments of Foreign Firms in China

Fig. 1. Foreign Firms in China.

Notes: This figure provides an overview of Foreign Investment Approval & Filing data (FIAP). In panel (a), we present the number of foreign firms in China each year from 1980 to 2018. Panel (b) shows the total amount of equity investment from foreign firms in China from 1980 to 2018.

3. Data and measurements

3.1. Data sources

This study is based on the joint availability of several newly constructed datasets, as explained below. In particular, our study requires two key measures to make the analysis possible. One is a reliable firm-level measure for corruption behavior. The other is detailed information on the shareholder list of foreign firms to capture the effect of the prevalent culture in a country of origin.

Annual Tax Survey (ATS). We utilize firm-level tax return data, which enables us to study both foreign and domestic firms, providing us with extensive and representative granular information at the firm level. In contrast to previous research that used data from the Chinese Industrial Enterprises database, we employ data from the Annual Taxation Survey (ATS). ATS data are obtained from surveys conducted annually by the Ministry of Finance and the State Administration of Taxation of China. Starting in 2004, the ATS has been implemented by local tax authorities. To ensure the quality of the survey data, the content of the survey was standardized, with detailed questions covering firms' information, including financial statements, tax standing, operational details, founding year, industry classification, and ownership information.³ Survey responses are also verified by local tax authorities by comparing them with the mandatory annual accounting reports of all Chinese firms, irrespective of listing status. These measures

³ ATS data only contain information on the type of the firm, such as state-owned, privately owned, or foreign-owned firms. We later complement ATS data using information from Foreign Investment Approval & Filing (FIAP) Data.

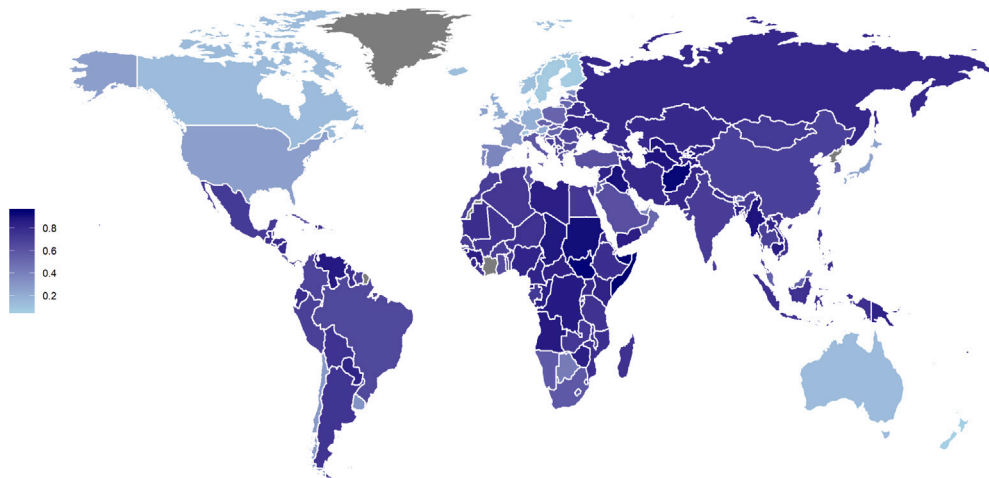


Fig. 2. Corruption index around the World.

Notes: This figure displays the average corruption index across countries worldwide, with a darker color indicates a more severe corruption. We construct the average corruption index based on the Corruption Perception Index (CPI) from Transparency International: we reverse defined the Corruption Perception Index (CPI), standardized and normalized it to get the country-year level corruption index which ranges from zero (very clean) to one (highly corrupt). We calculate the average corruption index for each country/area based on the entire sample period from 2004 to 2016.

lead to high accuracy and extensive coverage of the ATS data. In comparison to the extensively utilized Chinese Industrial Enterprises database, the ATS data demonstrates superior quality in terms of firm financial information. This dataset has been used in recent influential studies (Chen et al., 2021; Geng et al., 2022). For our research purposes, the ATS data contain information on firm financials, tax-related variables, and, importantly, entertainment costs, which enables us to measure firms' corruption behavior. We also utilize this data to study the economic consequences of corruption behaviors of multinationals, such as obtaining government subsidies and tax refunds, as well as the firms' profitability. Our ATS data cover the period from 2004 to 2016.

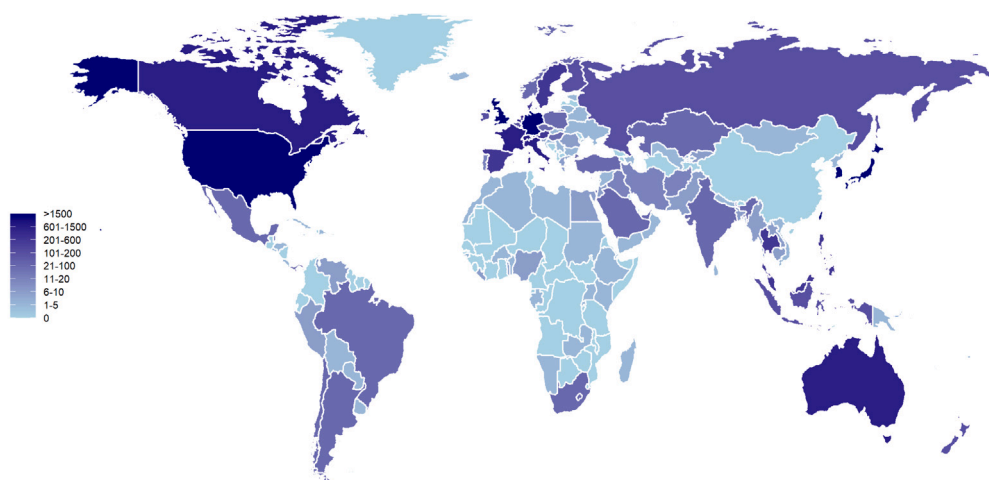
Foreign Investment Approval & Filing (FIAF) Data. The Foreign Investment Approval & Filing database (FIAF) originates from the initial reports submitted by foreign investors when establishing foreign-invested firms within China. It documents all foreign-invested firms that have obtained approval or filed records in China since 1980. Before 2016, foreign investors who established foreign-invested companies in China were required to go through approval processes. Starting in 2016, only filing was required. FIAF data record the equity investment information of all foreign firms in China, including the amount of each equity investment, the source country of the shareholder, and the changes in this information over time. These changes have led to variations in the source countries of equity investments and the amounts of equity investments, which are captured by the corruption index we constructed, as will be explained subsequently. Fig. 1 provides statistics from FIAF data on the number of foreign firms in China and their equity investments.

Corruption Perception Index (CPI). Our measure of corruption index is based on the Corruption Perceptions Index (CPI). The CPI is a metric that evaluates and orders countries based on the perceived extent of corruption within their public sectors. This index has been released yearly by the non-governmental organization Transparency International since 1995. In 2012 the methodology was modified, and we rescaled the index since 2012 to the range from zero to ten to ensure that they are comparable to the index before 2012. The higher the value, the more transparency it means. We reversely define this index so that a higher score represents higher levels of corruption and standardize and normalize it to the range from zero (very clean) to one (highly corrupt). We demonstrate significant variations in corruption levels across countries worldwide in Fig. 2: corruption levels are generally higher in African and Latin American countries, while corruption levels are typically lower in Western European and Australasian countries.

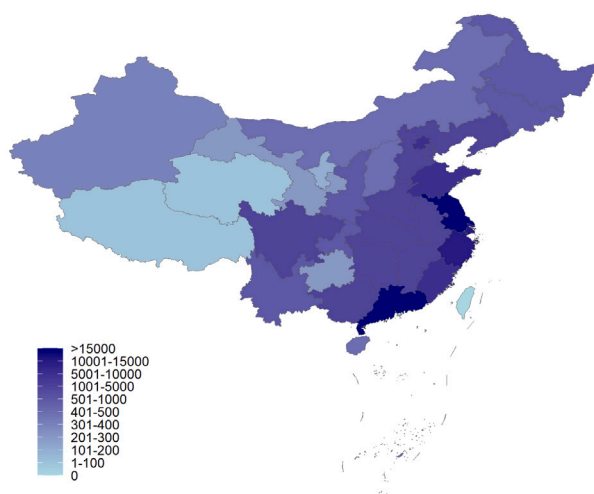
Other Data. To enhance the robustness of our empirical analysis, we have supplemented our primary data with multiple external sources. Specifically, we incorporated World Bank datasets to validate our firm-level corruption measures and acquired patent records from the Chinese Patent Office to examine innovation outcomes associated with corrupt practices. Additional datasets will be systematically introduced in subsequent sections as they become relevant to our analytical framework.

3.2. Sample and variables

The Sample. This paper uses the terms “foreign firms” and “multinational corporations (MNCs)” interchangeably. Our baseline sample consists of the foreign firm sample, using which we analyze the effect of the corruption culture of the source countries of foreign firms on their corruption behavior, as well as on the majority of other sections. We also employ a sample for domestic firms, which is used to study how the corruption culture of foreign firms spreads to domestic firms. The summary statistics for the foreign firm sample and the domestic firm sample are shown in Panel A and Panel B of Table 1, respectively. All continuous variables in Table 1 are winsorized at the 1% level before analysis. The foreign firm sample includes 91,895 firms across 123 source countries with 537,467 observations from 2004 to 2016. The domestic firm sample includes 308,428 firms with 1,660,365 observations from



(a) Foreign Firms in China by Countries of Origin



(b) Distribution of Foreign Firms in China

Fig. 3. Distribution of foreign firms.

Notes: This figure shows the spatial distribution of foreign firms in China from the perspective of their source countries and the distribution of foreign firms across provinces in China based on the Foreign Investment Approval & Filing data (FIAF) and Annual Tax Survey (ATS) data. A darker color indicates a larger number. Panel (a) shows the number of foreign firms established in China by countries of origin worldwide. Panel (b) shows the location by province of foreign firms in China.

the same period. Fig. 3 presents the geographical distribution of foreign firms in China based on our baseline sample, illustrating both their country of origin composition and provincial-level spatial allocation across China.

Figure A1 presents information on ownership changes in our foreign firm sample. We summarize the annual changes in ownership within our foreign firm sample (excluding the base year, 2004). Figure A1(a) shows the number of ownership changes each year, while Figure A1(b) illustrates the proportion of firms that experienced ownership changes relative to the total number of firms each year. The number of firms in the ATS data increased sharply from 2004 to 2005, leading to a significant increase in the number of firms included in our matched sample for that period. As a result, many firms in 2005 in our sample could not be compared with those from the previous year, resulting in a very small numerator. Consequently, the change ratio in 2005 was very low, but it rose sharply in subsequent years.

We also summarize the share of total assets according to whether their largest source of investment country is more or less corrupt than China. In our foreign firm sample, 99.180% of assets are from firms whose largest source of investment are from countries that are less corrupt than China, while 0.820% are from countries that are more corrupt than China.

Table 1

Summary statistics.

Panel A: Foreign Firm Sample						
Variables	N	Mean	SD	P25	P50	P75
EC	537,467	0.304	0.669	0.000	0.076	0.309
Corruption Index	537,467	0.364	0.172	0.225	0.334	0.517
Size	537,467	3.238	2.651	0.000	4.094	5.438
Leverage	537,467	0.564	0.329	0.325	0.554	0.769
Firm Age	537,467	10.003	5.360	6	9	13
Subsidy	537,467	0.143	0.558	0.000	0.000	0.000
Tax Refund	537,467	0.014	0.042	0.000	0.000	0.004
ROA	537,467	0.052	0.156	−0.006	0.024	0.092
Patent	537,467	0.957	14.426	0	0	0
Citation	537,467	0.255	1.197	0	0	0
Panel B: Domestic Firm Sample						
Variables	N	Mean	SD	P25	P50	P75
EC	1,660,365	0.319	0.671	0.000	0.064	0.342
Size	1,660,365	3.123	2.443	0.000	3.738	5.088
Leverage	1,660,365	0.681	0.317	0.480	0.720	0.901
Firm Age	1,660,365	11.508	7.306	6	10	14
Foreign Inflow ^{Corruption High}	1,660,365	0.899	1.924	0.000	0.000	0.000
Foreign Inflow ^{Corruption Low}	1,660,365	4.256	4.399	0.000	4.111	8.451

Notes: This table presents summary statistics of variables used in this study. Panel A shows the variables in the baseline sample of foreign firms in China. The sample is at the firm-year level and consists of 91,895 unique firms between 2004 and 2016. We construct the sample using the information on shareholder's country of origin from Foreign Investment Approval & Filing Database (FIAF) sourced from Ministry of Commerce of China and firm's financial and corruption behavior measure from Annual Tax Survey (ATS) Data obtained from the survey conducted by the Ministry of Finance and the State Administration of Taxation of China. Details of the sample construction can be found in Appendix B: Data Construction. Following (Cai et al., 2011) and Giannetti et al. (2021), we use firm's entertainment cost (EC) provided in ATS data as a proxy for firm's corruption behavior. EC is defined as the entertainment cost of a firm scaled by its total asset in a given year. To quantify a foreign firm's corruption culture, we define a firm level measure. Corruption Index_{*i,t*} = $\sum_{k=1}^K \left(\frac{\text{Amount}_{i,k,t}}{\sum_{k=1}^K \text{Amount}_{i,k,t}} \times \text{CPI}_{k,t} \right)$, where *k* indicates the *k*th shareholder of firm *i* in year *t*. Amount_{*i,k,t*} is the amount of investments from the *k*th shareholder who invest in firm *i* in year *t*. CPI_{*k,t*} denotes the Corruption Perception Index (CPI) of the country for the *k*th shareholder in year *t*. We source the Corruption Perception Index (CPI) from Transparency International and rescale the measure to a range between zero (very clean) and one (highly corrupt). We consider other firm variables including firm's size, leverage, age, and etc. All variables are defined in Appendix C: Variable Definitions. Panel B shows the variable of the panel of domestic firms, which consists of 308,428 distinct firms between 2004 and 2016. This sample is used for tests in Table 6 and only contains firms with consecutive two years of observations for our research purposes.

We merge the foreign firm part of ATS data with FIAF data to supplement the information about foreign firms in the ATS data. The matched observations between the foreign firm part of the ATS data and the FIAF data formed the baseline sample, the foreign firm sample of our study.

The foreign firm sample was constructed through the following steps. Firstly, we fill in the information for the years in the FIAF data in which the firms did not experience the ownership change, and we use the ownership information for the previous observation year to fill in the information for the years between that year and the year in which the next observation is located.

Second, in the FIAF data, we merge the time-varying CPI from Transparency International at the country level based on the source country for each equity investment of every foreign firm and the year of the equity investment. In this way, each observation corresponding to each equity investment is linked to a CPI. Then, we calculate our measure of corruption index for each foreign firm by using Eq. (1). As a result, we obtain our measure of corruption index for each foreign firm, taking into account the changes in shareholders and equity investments of each firm.

In the third step, we match the FIAF data with ATS data by firm name. We divide the ATS data into foreign firm and domestic firm based on firm type, and we extract the foreign firm part of ATS data for matching. The matching processes involve exact match and fuzzy match and the details can be found in Appendix B: Data Construction. We also manually check to ensure the accuracy of the matched results. Finally, after removing duplicate matches, we get a total number of 91,895 foreign firms successfully matched.

Finally, we append the result data from the third step with the domestic part of the ATS data to obtain the sample for analysis, which includes two types of firms: foreign firms and domestic firms. Corruption index is constructed by foreign firms. The final sample at the firm and year level consists of 2,197,832 observations with 537,467 observations of foreign firm across 123 source countries from 2004 to 2016 which is the foreign firm sample and 1,660,365 observations from 2004 to 2016 which is the domestic firm sample. Table A1 displays the industry distribution of firms in the sample of foreign firms.

Key Variables. Corruption behavior is an illegal act, making it difficult to measure based on objective data (Bertrand and Mullainathan, 2001). Since entertainment costs are directly taken from the firm's accounting records, they are more objective than survey data. Therefore, entertainment costs are widely used in research to measure corruption behavior (Cai et al., 2011; Huang et al., 2017; Giannetti et al., 2021). We employ entertainment costs (EC) as our primary measure of firm-level corruption behavior. In our baseline analysis, we construct the key dependent variable by scaling entertainment costs against firms' total assets. Panel A of Table 1 presents the descriptive statistics of EC in the foreign firm sample, with a mean of 0.304, a median of 0.076 and a standard deviation of 0.669. We also show descriptive statistics of EC in the domestic firm sample in Panel B of Table 1, with a mean of 0.319, a median of 0.064 and a standard deviation of 0.671.

In our baseline analysis, the key independent variable is the corruption culture of foreign firms. We have constructed a unique index to measure the culture of corruption at the firm level. This method takes into account the corruption culture of the source countries of shareholders in foreign firms, as well as the different amounts of the equity investments. Specifically, we calculate our measure of corruption index for each foreign firm by the following ways:

$$\text{Corruption Index}_{i,t} = \sum_{k=1}^K \left(\frac{\text{Amount}_{i,t,k}}{\sum_{k=1}^K \text{Amount}_{i,t,k}} \times \text{CPI}_{i,t,k} \right) \quad (1)$$

In Eq. (1), k indicates the k th shareholder of a firm i in year t . $\text{Amount}_{i,t,k}$ is the amount of investments from the k th shareholder who invest in firm i in year t . $\text{CPI}_{i,t,k}$ denotes the Corruption Perceptions Index (CPI) of the country for the k th shareholder in year t . We source the CPI from Transparency International and rescale the measure to a range between zero (very clean) and one (highly corrupt) before constructing our measure of corruption index. We show descriptive statistics of corruption index in Panel A of Table 1, with a mean of 0.364 and a standard deviation of 0.172.

Additionally, we use the difference in the corruption index between the two countries as an example to illustrate what one standard deviation of the corruption index actually means, and this is very helpful for us to interpret the reduced-form results. Panel A of Table 1 shows that one standard deviation of the firm-level corruption index is approximately 0.172. On average, this is roughly equivalent to the gap in corruption levels between China and South Korea. Therefore, if a firm's corruption index increases (or decreases) by one standard deviation, it means that, on average, the change in the firm's level of corruption is almost equivalent to the difference in corruption levels between China and South Korea. However, this does not imply that the level of corruption before and after an increase of one standard deviation in the firm's corruption index is equivalent to the corruption levels of South Korea and China. Instead, we are emphasizing that a change of one standard deviation in the firm's corruption index is roughly equivalent to the difference in corruption levels between China and South Korea.

Other Variables. To gain insights into the impact of China's legal frameworks and enforcement on the corruption behavior of foreign firm, we utilize the differences in the legal environments between Chinese provinces to analyze how province-level legal environments affect corruption behaviors of multinationals in China. The variable for legal index is based on China Market Index database (Wang et al., 2016) and it measures the development of market intermediary organizations and the legal integrity of the province in China where the firm is located, with higher values indicating a better legal environment in that province. Moreover, we also investigate the effect of corruption behavior on firm-level innovation. We construct two variables about the number of patent applications of firms each year and the average number of citations each year for all patents of a firm, respectively, based on the Chinese Patent data.

3.3. Validation of corruption measure

Due to its clandestine nature, identifying firm corruption through objective data is notoriously challenging. Consequently, much of the current research on corruption is based on subjective surveys at the individual level or on indices of perception of corruption specific to countries (Cai et al., 2011). We validate entertainment costs (EC) as a proxy for corruption behaviors by using other measurements of exposure to a corrupt environment to predict our EC measure. The assumption here is that when firms are exposed to a more corrupt environment, they are more likely to engage in corruption practices. Firms typically need to compete with other firms in the same industry or province for resources such as land, market share and policy incentives, so the characteristics at the industry or province level can effectively predict a firm's corruption behavior (Giannetti et al., 2021; Fang et al., 2023).

Table 2 shows the results of validation tests for our measure of corruption behavior, EC. We employ World Bank Enterprise Survey (2012) to measure exposure to a corrupt environment at province level. In columns(1)-(5), we consider five conventional proxies for exposure to a corrupt environment and examine the contemporaneous correlation between provincial-level corruption measures and EC, focusing on 2012. For example, we use percent of public transactions where a gift or informal payment was requested to measure the severity of corruption in Chinese provinces and test if it can predict our corruption behavior measure, EC. All results show that the extent that firms exposed to a corrupt environment is positively correlated with our proxy for corruption behavior, supporting EC as a valid measure for the corruption behavior of the firm.

4. Results

4.1. Corruption culture and corrupt behavior of multinationals

We first compare the average differences in corrupt behavior between foreign and domestic companies. We use EC to measure corruption behaviors of firms. Table A2 displays the differences in average levels of EC between foreign firms and domestic firms. We also demonstrate the changes over time in the EC of foreign firms and domestic firms, as shown in Figure A2. Overall, EC of foreign firms are lower compared to domestic firms.

Our main analysis below is based on the sample of foreign firms. We investigate the impact of corruption cultures from different source countries on the corruption behavior of multinationals in China to verify our hypothesis 1 (H1). To examine the relationship between corruption culture and corrupt behavior, we estimate the following baseline model:

$$EC_{i,t} = \beta \text{CorruptionIndex}_{i,t} + \mathbf{X}'_{i,t} \gamma + \delta_i + \tau_t + \epsilon_{i,t}, \quad (2)$$

Table 2
Validation tests for the corruption measure.

	Dependent Variable: EC				
	(1)	(2)	(3)	(4)	(5)
Bribery Incidence	0.374*** (0.045)				
Bribery Depth		0.422*** (0.048)			
Tax Bribery			0.467*** (0.042)		
Official Bribery				0.426*** (0.041)	
Corruption Constraint					0.288* (0.158)
Size	−0.011*** (0.002)	−0.011*** (0.002)	−0.011*** (0.002)	−0.011*** (0.002)	−0.010*** (0.002)
Leverage	0.029*** (0.010)	0.029*** (0.010)	0.029*** (0.010)	0.035*** (0.010)	0.030*** (0.010)
Firm Age	−0.000 (0.001)	−0.000 (0.001)	−0.000 (0.001)	−0.000 (0.001)	0.000 (0.001)
Observations	44,590	44,590	44,590	44,590	44,590
R-squared	0.034	0.035	0.036	0.036	0.033
Year FE	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES

Notes: This table presents the validation tests for the measure of corruption behavior. We consider five variables sourced from the World Bank Enterprise Survey (2012) that capture the features of local corruption as explained in the following: *Bribery Incidence* means percent of firms experiencing at least one bribe payment request; *Bribery Depth* means percent of public transactions where a gift or informal payment was requested; *Tax Bribery* means percent of firms expected to give gifts in meetings with tax officials; *Official Bribery* means percent of firms expected to give gifts to public officials to get things done; *Corruption Constraint* means percent of firms identifying corruption as a major constraint. The original data are based on firms from 25 representative cities in 2012. To increase the coverage of the tests, we follow (Giannetti et al., 2021) and aggregate the variables of interest to the province level. The original variables are presented as percentages and we multiply by 100 for each of these five variables. The sample used in this table is a cross-sectional snapshot in 2012 of the baseline sample of foreign firms described in Panel A of Table 1. We use these five corruption variables based on World Bank Enterprise Survey (2012) to predict firm's corruption behavior as captured by entertainment costs scaled by total assets (*EC*). Year fixed effects and industry fixed effects are included. The standard errors in parentheses are clustered at the firm level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

where $EC_{i,t}$ is the entertainment cost of firm i scaled by its asset in year t , measuring corrupt behaviors of firms. *Corruption Index* $_{i,t}$ measure corruption cultures of multinationals as constructed in Eq. (1). X_{it} is a set of time-varying firm controls including firm employment size, leverage ratio, and age. δ_i and τ_t are firm and year fixed effects, respectively. $\epsilon_{i,t}$ is the error term.

A key identification assumption of our specification is the conditional orthogonality of the corruption index and the error term. Our corruption index is constructed based on home country's culture and the equity share from each country. While the variation of home corruption culture is relatively exogenous, a legitimate concern is that multinationals may endogenously choose their locations in China based on the local corruption conditions, which may confound our estimate. Theoretically, as the investment of multinationals to start affiliates is on average large, the most important determinants for their location choice are supposed to be economic considerations, such as the market-orientation (horizontal and platform FDI), or cost-orientation (vertical FDI). Yet, it is possible that corruption culture might be a factor as well. If corruption culture is important in multinationals' decisions, we would expect that multinationals from more corrupt countries are more likely to locate in regions with higher corruption culture in China.

To test whether this concern is relevant, we regress the average corruption index of newly entered foreign firms on the lagged corruption culture at prefecture level, measured as the average *EC* of local domestic firms. The estimated coefficient, reported in column (1) of Table A3, is statistically insignificant. The result remains consistent with an alternative measure of local corruption culture, defined as the weighted average of the ratio of entertainment cost to total assets of local domestic firms in the previous year, as shown in column (2). Overall, these results indicate that the local corruption culture is not an important determinant of the locations of MNCs from different corruption cultures.

Baseline Results. Table 3 presents the estimation results of our baseline model examining the impact of corruption culture on foreign firms' corrupt behaviors. In successive model specifications, we progressively incorporate additional firm-level control variables. In columns (1) and (2), we only include year fixed effects. Column (2) includes firm-level characteristics. Specifically, Size is represented by the number of employees in the firm and received logarithmic transformation, Leverage is the ratio of debt to asset, and Firm Age refers to the age of the firm. In columns (3) and (4), we include year fixed effects, province fixed effects and industry fixed effects, while column (4) includes firm-level characteristics that is the same as column (2). In columns (5) and (6), we further control for year fixed effects and firm fixed effects.

As shown in column (4), when industry, province, and time fixed effects included, one standard deviation increase in corruption culture of multinationals corresponds to a 4.187% increase relative to the mean of the dependent variable. Column (6) presents the results of our most rigorous model specification, which includes year fixed effects and firm fixed effects and time-varying firm level characteristics. The coefficient presented in column (6) indicates that one standard deviation increase in corruption culture is associated with a 2.320% increase relative to the mean of *EC*.

Table 3
Corruption culture and corruption behavior of multinationals.

	Dependent Variable: EC							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Corruption Index	0.024*** (0.008)	0.027*** (0.008)	0.071*** (0.008)	0.074*** (0.008)	0.042** (0.017)	0.041** (0.017)	0.032* (0.018)	0.030* (0.018)
Size		−0.008*** (0.001)		−0.004*** (0.001)		0.021*** (0.001)		0.021*** (0.001)
Leverage		0.033*** (0.005)		0.040*** (0.004)		−0.021*** (0.007)		−0.021*** (0.007)
Firm Age		−0.004*** (0.000)		−0.004*** (0.000)		−0.007*** (0.002)		−0.007*** (0.002)
Corruption Index × Rent-Seeking Industry							0.224*** (0.055)	0.239*** (0.055)
Rent-Seeking Industry							−0.087*** (0.026)	−0.098*** (0.026)
Observations	537,467	537,467	535,695	535,695	537,467	537,467	537,467	537,467
R-squared	0.043	0.045	0.069	0.070	0.440	0.440	0.440	0.440
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Province FE	NO	NO	YES	YES	NO	NO	NO	NO
Industry FE	NO	NO	YES	YES	NO	NO	NO	NO
Firm FE	NO	NO	NO	NO	YES	YES	YES	YES

Notes: This table presents the baseline estimation of the impact of corruption culture of the country of origin on corruption behavior of multinationals in China. Dependent variable, *EC*, stands for the firm level entertainment costs scaled by total assets sourced from Annual Tax Survey (ATS). To quantify a foreign firm's corruption culture, we define a firm level measure. $CorruptionIndex_{i,t} = \sum_{k=1}^K (\frac{Amount_{i,k,t}}{\sum_{k=1}^K Amount_{i,k,t}} \times CPI_{k,t})$, where k indicates the k_{th} shareholder of a firm i in year t . $Amount_{i,k,t}$ is the amount of investments from the k_{th} shareholder who invest in firm i in year t . $CPI_{k,t}$ denotes the Corruption Perception Index (CPI) of the country for the k_{th} shareholder in year t . We source the Corruption Perception Index (CPI) from Transparency International and rescale the measure to a range between zero (very clean) and one (highly corrupt). We control for firm variables including firm size, leverage, and age. In columns (1) and (2), we only consider year fixed effects, while we further include province fixed effects and industry fixed effects in columns (3) and (4). We include year fixed effects and firm fixed effects in columns (5) and (6). Column (7) and (8) presents the heterogeneous effect of corruption culture on corruption behavior of firms conditional on the high rent-seeking industries. *Rent-Seeking Industry* is set to one if the firm is operating in a high rent-seeking industry as defined in Xu and Yano (2017), and zero otherwise. Standard errors in parentheses are clustered at the firm level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

The estimated coefficients in columns (1)–(6) remain remarkably stable and statistically significant. The findings reveal a robust positive association between the corruption culture of the source countries and the corruption behaviors of the firms. In other words, if the corruption issues in the home countries of multinationals are severe, these multinationals are more likely to engage in corruption behaviors in China as well. In conclusion, the empirical results support hypothesis 1 (H1).

Heterogeneity. We also extend our analysis by exploring the heterogeneous effects of corruption culture according to rent-seeking activities in a given industry. High rent-seeking industries in China consist primarily of strategically significant sectors, often characterized by concentrated state ownership and strong ties to the government. Firms operating within these industries, marked by the presence of monopoly rents, are further motivated to focus on rent-seeking behaviors to extract additional resources and vie for increased profits. In these high rent-seeking industries, obtaining favoritism from officials through corrupt behaviors is essential for the survival and growth of firms (Zhang et al., 2019). Based on this information, we define high rent-seeking industries according to Xu and Yano (2017). In this context, high rent-seeking industries mainly include mining, public administration and social security, and the real estate sector. By interacting a dummy variable for the high rent-seeking industry with the $CorruptionIndex_{i,t}$, we rerun the following regression:

$$EC_{i,t} = \beta_1 CorruptionIndex_{i,t} + \beta_2 CorruptionIndex_{i,t} \times Rent - SeekingIndustry_{i,t} + \beta_3 Rent - SeekingIndustry_{i,t} + \mathbf{X}'_{i,t} \boldsymbol{\gamma} + \delta_i + \tau_t + \epsilon_{i,t} \quad (3)$$

Columns (7)–(8) of Table 3 show the heterogeneity test on whether an industry is high rent-seeking or not. In high rent-seeking industries, multinationals from more corrupt countries have higher EC, which shows that these multinationals engage in more corrupt behaviors.

Robustness. We conduct a variety of robustness checks. First, a concern is that corruption culture may mainly reflect the economic performance of the country, which can confound the interpretation. We replace Corruption Perceptions Index (CPI) with the log of GDP per capita in Eq. (1) to construct a placebo measure of corruption culture and rerun specification (2). Results in Table A4 show that this placebo measure is uncorrelated with the corrupt behaviors of MNCs in China, indicating that our measure does not only capture the economic condition of the home country, and our finding is robust to this concern.

Second, we consider alternative model specifications in our baseline model. We introduce controls for industry-year fixed effects and province-year fixed effects, respectively. Table A5 presents results based on the alternative model specifications. Column (1) includes industry-year fixed effects, column (2) includes province-year fixed effects, and column (3) includes both of these fixed effects. In columns (1)–(3), other model specifications are consistent with the baseline model, and the results remain robust.

Third, we consider the alternative measure of dependent variable in our baseline estimation. Table A6 presents results based on the alternative measure of our dependent variable, EC. We use revenue of the firm to scale the entertainment cost and obtain our alternative measure, EC2. Columns (1)–(2) follow the same model specifications as the baseline setting in columns (5)–(6)

of Table 3. The baseline results are insensitive to the alternative measure. The estimated coefficients in columns (1)–(2) remain remarkably stable and statistically significant at the 1% level. Additionally, our findings are also robust to alternative samples. We present in columns (1) and (2) of Table A7 the robustness check to the baseline estimation by limiting observations related to the wholly-owned foreign firms. We also present in column (3) and (4) of Table A7 the robustness checks to the baseline estimation by drop observations related to the tax haven countries. Our baseline results still hold under these alternative samples. Besides, we also randomly drop 5% of the observations in our foreign firm sample, estimate t-values for the coefficient of interest, and repeat this process 500 times. We summarize the results in Figure A3.

Fourth, we additionally control firm performance in our baseline regression. We use the wage and net income as separate proxies for firm performance. Table A8 presents robustness checks to the baseline estimation by including firm-level performance as an additional control. Columns (1) to (2) include wage, while columns (3) to (4) include net income. The baseline results are still robust after controlling the firm performance.

4.2. Economic consequences

Having established that MNCs from more corrupt home countries are more likely to engage in corrupt activities when operating in China, we now turn to the economic consequences of these behaviors on MNCs. In this subsection, we investigate the impact of EC on various aspects of firm performance. Specifically, we first examine whether companies that engage in corruption effectively secure preferential treatment from the government. Next, we analyze the effects on firms' profitability and innovation.

Subsidy and Tax. In China, to support the development of specific industries, the government provides a range of economic incentives to firms within those industries. This typically includes government subsidies and tax exemptions, aimed at strategically developing certain sectors and promoting innovation activities within these firms. Ideally, these measures can improve resource allocation and promote the development of specific industries, such as the high-technology industry. However, public officials are often self-serving, and in situations where the government monopolizes the supply of public goods, these officials may abuse their positions for personal gain. Such behaviors include soliciting bribes to provide subsidies, issuing permits, embezzling public funds, and selling government-owned goods at black market prices (Alm et al., 2016). For instance, in China, the authority to grant subsidies lies with government officials, creating an incentive for firms to engage in corruption behaviors by offering bribes to local government officials in exchange for subsidies (Fang et al., 2023). A higher level of corruption in society can also lead to more tax evasion, as corrupt officials seek additional income through bribery (Banerjee et al., 2022).

In this subsection, we first explore the returns that corruption culture and corruption behavior by multinationals can yield from government officials. We assess the association between corruption culture of firms and their potential economic returns from government. We consider two types of government concessions: subsidy and tax refund. The model is shown in Eq. (4), and we use subsidy and tax refund to represent government concessions, respectively. For each kind of government concessions, our model follows the baseline model with firm fixed effects, year fixed effects and firm-level controls.

$$y_{i,t} = \beta EC_{i,t} + \mathbf{X}'_{i,t} \gamma + \delta_i + \tau_t + \epsilon_{i,t} \quad (4)$$

In Eq. (4), $EC_{i,t}$ is the entertainment cost of foreign firm i scaled by asset in year t . $y_{i,t}$ are the outcomes that we are interested in, including the subsidy and tax refund at the firm-year level. We employ the foreign firm sample to run the regressions. The results are presented in columns (1) and (2) of Table 4.

As shown in column (1) of Table 4, one standard deviation increase in EC is associated with a 0.017 increase in the government subsidy that is assigned to the firm, which corresponds to 2.997% of the standard deviation of the dependent variable, subsidy. The impact of EC on government subsidy suggests that multinationals engaging in more corruption behaviors received more government subsidies in China. In column (2), a standard deviation increase in EC is associated with a 0.001 increase in the tax refund that is returned to the firm, which corresponds to 3.186% of the standard deviation of the dependent variable, tax refund. The impact of EC on tax refund suggests that multinationals engaging in more corruption behaviors received more tax refund from Chinese government.

Profitability. The increased EC of MNCs has “benefited” them with more government preferential treatments. These advantages may provide firms with a competitive edge and seemingly enhance their market position. If that is the case, these corrupt firms may acquire larger market shares and profits, which may distort the resource allocation (Hang and Zhan, 2023). We therefore examine whether the profitability of firms is related to their EC behaviors. We calculate return on asset, $ROA_{i,t}$, by dividing a firm's net income by its total assets to gauge the profitability of a firm. Using $ROA_{i,t}$ as the dependent variable to rerun regression (4), we obtain the results presented in column (3) of Table 4.

We find that one standard deviation increase in EC is associated with a 0.007 increase in the ROA of the firm, which corresponds to 4.487% of the standard deviation of ROA. The results indicate that multinationals engaging in more corrupt behaviors have better profitability in China.

Innovation. Innovation is crucial for the success of firms and serves as the ultimate driver of economic growth. Alleviating financial constraints by increasing profitability can significantly benefit firms' innovation activities. However, if firms can rely on corrupt practices rather than investing in innovative efforts to maintain their competitive edge and realize higher profitability, firms' incentives to innovate may be dampened.

We develop two measures to assess how firm innovation is affected by EC: $Patent_{i,t}$ as the number of patent applications by firm i in year t , and $Citation_{i,t}$ as the average citations for patents of firm i in year t to account for innovation quality. The results are

Table 4
Economic consequences.

	Dependent Variable				
	Subsidy	Tax Refund	ROA	Patent	Citation
	(1)	(2)	(3)	(4)	(5)
EC	0.025*** (0.002)	0.002*** (0.000)	0.010*** (0.001)	−0.018 (0.039)	−0.006 (0.012)
Size	0.009*** (0.001)	0.001*** (0.000)	−0.000 (0.000)	0.035* (0.021)	0.049*** (0.009)
Leverage	−0.007 (0.005)	−0.000 (0.000)	0.018*** (0.002)	0.066 (0.103)	−0.000 (0.039)
Firm Age	−0.008*** (0.002)	−0.000*** (0.000)	−0.004*** (0.001)	0.002 (0.036)	0.021 (0.023)
Observations	537,467	537,467	537,467	537,467	537,467
Observations (Drop singletons and separations)				129,710	85,036
R-squared	0.499	0.553	0.549		
Pseudo R2				0.694	0.278
Year FE	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES

Notes: This table presents how foreign firms' corruption behavior affect government concessions and their economic and innovation outcome. *EC* is defined as entertainment costs scaled by total assets in a given year. *Subsidy* is defined as the government subsidy received by a firm scaled by its total asset. *Tax Refund* is defined as the tax refund received by a firm from government scaled by its total asset. *ROA* is defined as the net income of a firm divided by its asset. *Patent* is defined as the number of patent applications of the firm each year. *Citation* is defined as the average number of citations per year for all patents of a firm. The model includes firm and year fixed effects, and we control for firm variables including firm size, leverage, and age. Column (4) and (5) are estimated by Poisson pseudo-maximum likelihood method, which is superior in cases where the outcome variable is count-based or shows many zero values (Cohn et al., 2022). Standard errors in parentheses are clustered at the firm level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table 5
Enforcement and anti-corruption campaign.

	Dependent Variable: EC					
	(1)	(2)	(3)	(4)	(5)	(6)
Corruption Index	0.125*** (0.032)	0.118*** (0.031)				
Corruption Index × Legal Index	−0.013*** (0.004)	−0.012*** (0.004)				
Legal Index	0.015*** (0.002)	0.015*** (0.002)				
Corruption Index ₂₀₁₁ × Post_AntiCorruption			−0.047*** (0.013)	−0.045*** (0.013)		
Corruption Index ₂₀₁₁ × Post1					−0.072*** (0.014)	−0.071*** (0.014)
Corruption Index ₂₀₁₁ × Post2					−0.060*** (0.015)	−0.059*** (0.015)
Corruption Index ₂₀₁₁ × Post3					−0.083*** (0.017)	−0.082*** (0.017)
Corruption Index ₂₀₁₁ × Post4					−0.027 (0.017)	−0.026 (0.017)
Observations	537,467	537,467	415,288	415,288	415,288	415,288
R-squared	0.440	0.440	0.404	0.404	0.404	0.404
Controls	NO	YES	NO	YES	NO	YES
Year FE	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES

Notes: This table presents the effect of Law enforcement and Chinese anti-corruption campaign on corruption behavior of foreign firms, respectively. *Legal Index* measures the development of market intermediary organizations and the legal integrity of the province in China where the firm is located, with higher values indicating a better legal environment in that province. The legal index is based on the China Market Index database (Wang et al., 2016). *CorruptionIndex₂₀₁₁* is the corruption index of the sample firm in 2011. *Post_AntiCorruption* is set to one for years after 2011, and zero otherwise. *Post_{t,w}* is an indicator set to one if with year relative to 2011, the year preceding the enforcement of the anti-corruption campaign in China. Other model specifications follow the conventions in Table 3. Standard errors in parentheses are clustered at the firm level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

reported in columns (4) and (5). We find no significant effect of EC on firms' innovation, although these firms' profitability has increased, as indicated in column (3).

Overall, these findings indicate that MNCs can derive considerable benefits from corrupt activities by securing preferential government treatment, which contributes to their enhanced profitability. However, this advantage does not extend to fostering innovation, an essential driver of long-term economic growth. The results also suggest that MNCs who devote fewer resources to corruption may pose disadvantages in receiving subsidies, tax refunds, and other preferential treatment from the government, and consequently are less profitable. Combining the baseline findings that MNCs engage in elevated levels of corrupt practices in high rent-seeking industries in particular, these results raise concerns on the economic costs of corruption.

4.3. Enforcement

Considering the above economic consequences, how can the corrupt activities of MNCs in the host country be curtailed? A substantial amount of research emphasizes the role of law enforcement in shaping economic activities such as corruption (Allen et al., 2005; Lu et al., 2015; Zilibotti, 2017) and pollution behavior (Kong and Liu, 2024). In this section, we explore the impact of China's legal frameworks and enforcement on the corruption behavior of foreign firms to test our hypothesis 2 (H2). We first utilize the differences in the legal environments between Chinese provinces to analyze how difference in province-level legal environments affect corruption behavior of multinationals in China. Furthermore, we investigate the changes in corruption behavior of multinationals in China following the anti-corruption campaign.

Legal Environment. Despite China's economy transitioning towards a more market-oriented way, local governments in China still dominate resource allocation through mechanisms such as operating permits, entry controls, and price regulations (Chen et al., 2011). In China, local governments bear comprehensive responsibilities within their jurisdictions for initiating and coordinating reforms, providing public services, and formulating and enforcing laws. Across different provinces, the formal institutions such as regulatory bodies and law enforcement agencies vary significantly in the degree of protection they offer to the property rights of private firms. These differences present a rare opportunity for us to examine whether variations in legal protections within a country lead to differences in corruption behavior among foreign firms.

We utilize the difference in legal environment across various provinces in China. We employ the "Development of market intermediary organizations and the legal institutional environment" index from the China Market Index Database (Wang et al., 2016) to measure the inter-provincial variations in the legal protection of market operations. Based on this index, we construct a continuous variable, $Legal\ Index_{i,t}$, where a higher value indicates that the provincial government has more comprehensive legal enforcement for protecting marketization. We then interact it with $Corruption\ Index_{i,t}$ to assess how local legal environment may curb the transmission of home corruption culture on corrupt behaviors of affiliates in

As shown in columns (1)-(2) of Table 5, the positive impact of corruption culture on EC in the baseline estimation turns negative when accounting for differences in the legal environment. In provinces with better legal environments, EC decreases more, reflecting the inhibitory effect of a sound legal environment on corruption.

The Anti-Corruption Campaign in China. After the 18th National Congress of the Communist Party of China in 2012, President Xi Jinping launched a far-reaching anti-corruption campaign that targeted not only high-ranking government officials but also grassroots cadres. Starting in May 2013, the Central Commission for Discipline Inspection dispatched several central inspection teams to investigate corruption behaviors among officials in different provinces and some state-owned enterprises. Subsequent inspections were carried out two or three times each year to thoroughly examine the behavior of officials or senior executives of state-owned enterprises (Hao et al., 2020). This anti-corruption campaign was more intense and enduring than previous ones. In just one year, 2016, over 100,000 low-ranking officials were prosecuted for corruption, compared to 30,315 in 2012 (Chen and Kung, 2019).

In this section, we examine the impact of China's anti-corruption campaign on corruption behavior of multinationals. To identify the impact of the anti-corruption campaign on the corruption behavior of multinationals in China, we adopt a Difference-in-Differences (DID) method to address potential endogeneity concerns:

$$EC_{i,t} = \beta CorruptionIndex_i^{2011} \times Post_Anticorruption_t + \delta_i + \tau_t + \epsilon_{i,t}, \quad (5)$$

where $EC_{i,t}$ is the entertainment cost of firm i scaled by asset in year t , and we employ foreign firm sample to do this analysis. $Corruption\ Index^{2011}_i$ is the corruption index that we constructed of firm i in 2011. $Post_Anticorruption_t$ is a dummy set to one for years after 2011, and zero otherwise. We also conduct a dynamic version of Eq. (5), in which we interact the corruption index of 2011 with time dummies as follows:

$$EC_{i,t} = \beta_c \sum_{c=-4}^4 Post_c \times CorruptionIndex_i^{2011} + \delta_i + \tau_t + \epsilon_{i,t}, \quad (6)$$

where $Post_c$ is a series of year dummies, which means c years relative to 2011, the year preceding the enforcement of the anti-corruption campaign in China, and we consider 4 years period before and after 2011. Columns (3)-(4) of Table 5 show the estimation results of Eq. (5). We find a significant decrease in EC among multinationals in China after the launch of the anti-corruption campaign. We plot the estimated coefficients of β_c of Eq. (6) in Fig. 4 and report the results in columns (5)-(6) of Table 5. We find that after the anti-corruption campaign was launched, the coefficient changed from being positively significant to negatively significant, suggesting a deterring effect of Chinese anti-corruption campaign on multinationals in China. In a word, these results confirm the role of China's legal framework and law enforcement in curbing corruption behavior, supporting our hypothesis 2 (H2).

4.4. Cultural spillover

Multinationals are not only engines of economic growth and development, but also a link in the spread of ideas and culture (Alfaro, 2017; Tang and Zhang, 2021). In this section, we examine whether and how multinationals spread the home country corruption culture to local firms of the host countries to verify our hypothesis 3 (H3).

We first categorize the MNCs into two types based on whether the firm corruption index is higher than China's Corruption Perceptions Index (CPI). We measure the inflows of these two types of MNCs entry of foreign firms into China at the city and year level in China, specifically:

$$ForeignInflow_{c,t}^{Corruption\ High} = \log \left(\sum_{i \in \{i|h(i)>h(\text{China})\}} Employee_{i,c,t} \right), \quad (7)$$

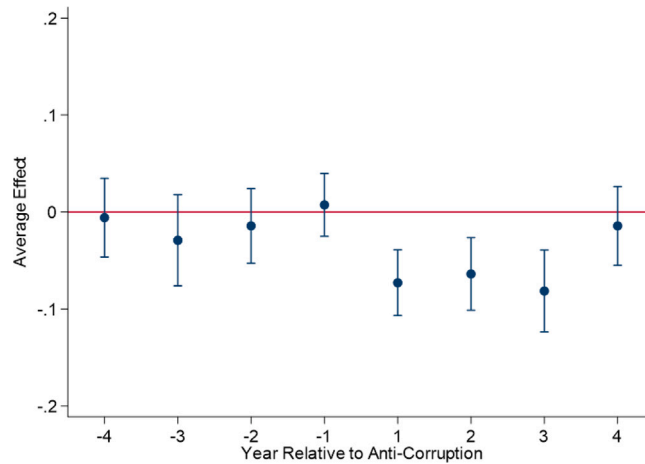


Fig. 4. Dynamic effect of anti-corruption.

Notes: This figure shows the dynamic effect of Chinese anti-corruption campaign based on coefficients as reported in column (5) and (6) of Table 4 using the foreign firm sample reported in Panel A of Table 1. We use 2011, the year preceding the enforcement of the anti-corruption campaign in China, as the baseline year, which stands for period 0. We considered the four years before and after the baseline year and generate dummy variables, respectively. We created interaction terms by taking the corruption index in 2011, the year before the anti-corruption campaign, and each year dummy variable to run Eq. (6).

where $h(i)$ denotes the corruption index of the firm i , $Employee_{i,c,t}$ is the number of employees from the newly established foreign firm i operating in city c in year t . $ForeignInflow_{c,t-1}^{Corruption\ High}$ captures the inflow of foreign firms from countries more corrupt than China. Conversely, we define $ForeignInflow_{c,t}^{Corruption\ Low}$, which measures the inflow of foreign firms from countries less corrupt than China. The inflow variables are lagged one year to capture the transmission effect on the corruption behavior of domestic firms in a lead-lag fashion.

We follow Eq. (8) to test the transmission of corruption culture from foreign firms to domestic firms, specifically:

$$EC_{i,t} = \beta_1 ForeignInflow_{c,t-1}^{Corruption\ High} + \beta_2 ForeignInflow_{c,t-1}^{Corruption\ Low} + \delta_i + \tau_t + \epsilon_{i,t}, \quad (8)$$

where $ForeignInflow_{c,t-1}^{Corruption\ High}$ captures the inflow of foreign firms from countries more corrupt than China and is lagged one year to capture the transmission effect on the corruption behavior of domestic firms in a lead-lag fashion. $ForeignInflow_{c,t-1}^{Corruption\ Low}$ captures the inflow of foreign firms from countries less corrupt than China and is lagged one year to capture the transmission effect on the corruption behavior of domestic firms in a lead-lag fashion.

We use domestic firm sample to estimate this model, and the regression results are reported in column (1) and (2) of Table 6. We first include only year and firm fixed effects in column (1) and add firm-level controls include size, leverage and firm age in column (2). The coefficient presented in column (2) indicates that one standard deviation increase in corruption culture inflow from multinationals in which the corruption culture is lower than corruption culture in China is associated with a 0.018 decrease in the corruption behaviors of domestic firms located in the same city as these foreign firms, which corresponds to 5.792% of the mean of the dependent variable, EC. We also consider an alternative measure of independent variables, $ForeignInflow_{c,t-1}^{Corruption\ High}$ and $ForeignInflow_{c,t-1}^{Corruption\ Low}$. We replace employee with asset of firm to construct the independent variables by the same procedure, and the regression result is shown in column (3) and (4) of Table 6. The transmission effect of corruption culture inflow from multinationals with lower corruption index remains robust with this alternative variable. These findings also suggest that the observed changes in domestic firms' behaviors are driven more by cultural spillovers than by economic conditions alone.⁴

Overall, our findings indicate that the culture of multinational corporations with a lower level of corruption than that of China plays a significant role in improving the corruption environment for China's domestic firms. The empirical results in this section support our third hypothesis (H3).

5. Conclusion

Multinationals are believed to have a close relationship with the global spread of corruption practices; however, the specific impact of corruption norms on corruption behavior has not yet been thoroughly researched. In this work, we utilize administrative data and tax returns of multinational corporations (MNCs) to investigate the impact of foreign cultural norms on corrupt practices within China. Leveraging time-varying ownership details of foreign shareholders, our work suggests that multinationals with more

⁴ If domestic firms' behaviors were driven solely by economic conditions, then MNCs from both more corrupt and less corrupt cultures would have similar rather than different impacts on local firms' behaviors.

Table 6
Cultural transmission.

	Dependent Variable: EC			
	Inflow Constructed by Employees		Inflow Constructed by Assets	
	(1)	(2)	(3)	(4)
Foreign Inflow ^{Corruption High}	0.0008** (0.0003)	0.0008** (0.0003)	0.0001 (0.0001)	0.0001 (0.0001)
Foreign Inflow ^{Corruption Low}	−0.0041*** (0.0003)	−0.0042*** (0.0003)	−0.0009*** (0.0002)	−0.0010*** (0.0002)
Size		0.0220*** (0.0007)		0.0220*** (0.0007)
Leverage		−0.0059* (0.0034)		−0.0053 (0.0034)
Firm Age		−0.0008 (0.0008)		−0.0009 (0.0008)
Observations	1,660,365	1,660,365	1,660,365	1,660,365
R-squared	0.4740	0.4746	0.4739	0.4745
Year FE	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES

Notes: This table shows the effect of foreign firm inflow on the corruption behavior of Chinese domestic firms. The results are based on the panel of domestic firms as described in Panel B of Table 1. We define the inflow of foreign firms using two separate variables based on the corruption culture of their source countries relative to China. In column (1) and (2), we construct foreign firm inflow based on number of employees of the foreign firm. $\text{Foreign Inflow}_{c,t}^{\text{Corruption High}} = \log \left(\sum_{i \in \{i|h(i) > h(\text{China})\}} \text{Employee}_{i,c,t} \right)$, where $h(i)$ denotes the corruption index of the firm i , $\text{Employee}_{i,c,t}$ is the number of employees from the newly entered foreign firm i operating in city c in year t . $\text{Foreign Inflow}_{c,t}^{\text{Corruption High}}$ captures the inflow of foreign firms from countries more corrupt than China. Conversely, we define $\text{Foreign Inflow}_{c,t}^{\text{Corruption Low}}$. The inflow variables are lagged one year to capture the transmission effect on the corruption behavior of domestic firms in a lead-lag fashion. In Columns (3) and (4), we repeat the process of constructing these variables, based on the total assets of foreign firms, and then repeat the regression. Standard errors in parentheses are clustered at the firm level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

corrupt culture in the home country conduct more corruption behaviors in China, as evidenced by increased entertainment costs. The effect is more pronounced in high-rent-seeking industries. We also document the economic consequences of corruption behavior of MNCs including receiving more subsidies and tax refunds from local governments and having higher profitability, but we find no evidence to support the relationship between corruption behavior and firm innovation. We explore the effect of regulatory enforcement in curbing corruption behavior and find that China's anti-corruption campaign has significantly reduced the influence of foreign corruption culture, emphasizing the importance of robust regulatory enforcement in shaping the behavior of foreign firms operating in emerging economies. Our results also highlight the spillover effect of the corruption culture of multinationals on domestic companies in China. We find that multinationals with a lower level of corruption than China can lead to a reduction in corruption behaviors among Chinese domestic firms located in the same city. Our findings provide new evidence linking the corruption culture of multinationals' home countries to the corruption behavior of multinationals in host countries and the corruption behavior of domestic firms in those host countries.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jebo.2025.107012>.

Data availability

Data will be made available on request.

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